



A Framework for AI-Assisted Storytelling from Collective Multimedia Records

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Abstract. Digital Storytelling (DS) is an evolving and fundamental form of narrative that is crucial for cultural expression and collaborative content creation, particularly on web-based platforms. Despite the potential of modern co-creation environments, such as the Polariscope platform, users often struggle with developing full, high-quality narratives due to constraints in time, patience, or creative confidence. Addressing this critical barrier, this research explores the integration of Generative Artificial Intelligence (GenAI) as an assistive technology to optimize and facilitate the creative workflow. This research focuses on integrating GenAI into the Polariscope platform, aiming to enrich the creation and co-creation of stories, as well as content metadata and recommendations. It is intended to develop and analyze an intelligent assistant capable of supporting users in writing, generating descriptions for content, and automatically creating stories, utilizing the platform's existing content. Overall, the study aims to investigate the extent to which Artificial Intelligence (AI) can enhance the creativity, efficiency, and quality of generated stories, as well as to assess user interest and engagement with this type of creative assistance.

Keywords: Artificial Intelligence, Co-creation, Digital Storytelling, Creativity, Stories, Intelligent Assistant.

1 Introduction

The Polariscope is a Research and Development (R&D) project focused on establishing a collaborative digital storytelling platform. Its primary goal is to bring together citizens and cultural heritage institutions to share archives and memories, thereby promoting their co-creation and visualization. To enhance this process, the project proposes an AI-assisted solution capable of supporting users in the creation of narratives and automatically generating stories based on existing content. In the platform's context, a story is part of the data aggregation and organization hierarchy, which is structured into three levels: Projects, Stories, and Content. The highest level is the Project (e.g., a collective cultural event, such as a national Independence Day celebration). Within each Project we have the Stories, which are the narratives designed by the creators, encompassing

text, content, and memories; this is the level where the AI assistant is designed to provide support. At the base are the Contents, which include photos, videos, audio files, or citations that users share, and which can later be used within the Stories.

In this context, a critical challenge is the creation of Stories. Based on previous testing [1][2], it has been observed that most users tend to prefer either posting content or simply viewing what others have published. This behavior imposes significant limitations for the use of Polariscope, as the story creation is a longer process demanding more attention, time, and creativity. This possible limitation establishes the basis for the current research. Given the significant evolution of GenAI in recent years and the ongoing emergence of various tools supporting digital storytelling, there is a clear opportunity to use GenAI to assist creators in the co-creation of stories, recommending content, aiding in narrative writing, or even generating stories automatically.

The study is guided by the core research question: “Can the GenAI assistant significantly contribute to the story creation process and to its perceived quality?” This is encapsulated in the general objective of the research of developing a tool for the Polariscope platform to ease and improve the process of creating stories, through the integration of AI. This research adopts the Design-Based Research (DBR) methodology. This means the study is focused not merely on technological development, but on the iterative design and practical improvement of the assistant, emphasizing the user experience to ensure the process is made as easy and intuitive as possible for the story creators.

This paper is structured into six sections: the first establishes the research context; the second presents insights from previous work; the third provides an overview of the state of the art; the fourth details the methodology and its phases; the fifth outlines the proposed solution; and the final section provides a summary of the study and directions for future work.

2 Previous Work

Key findings from prior research [1][2] related to AI-assisted storytelling conducted within the Polariscope R&D project established relevant insights. In this preliminary research, the team aimed to find the optimal value of the AI’s Temperature parameter, which governs its level of creativity. The findings established that this parameter cannot be fixed. In these tests employing the GPT 3.5 model, users demonstrated varying preferences that were directly linked to the AI’s output creativity. Some participants favored stories generated with a high Temperature, indicating a preference for more creative and engaging narratives. Conversely, other participants preferred stories generated with a low Temperature, resulting in more formal and objective texts. We hypothesize that this divergence is driven by the fact that different narrative contexts demand distinct levels of factual adherence. For instance, a story about a local festive event, which often appeals to personal sentiment, showed a clear preference for the higher-temperature texts. Conversely, a narrative concerning a historical event, which inherently requires rigor and objectivity, led users to prefer the lower-temperature text.

This discovery directly informs the design of the AI assistant and poses the fundamental challenge that this research must address: how will the system dynamically decide the appropriate level of AI creativity within the context of the Polariscope platform? This issue is critical, as a one-size-fits-all approach to generative output would fail to meet user expectations across the varied story types within the platform's archives. This challenge directly shapes the requirements and evaluation metrics for the intelligent assistant proposed in this paper.

3 State of the Art

This work aims to develop an intelligent tool within the Polariscope platform to ease and enhance the story creation process through the integration of AI. To support this objective and the conceptual and theoretical foundation a literature review was prepared around three core pillars:

- **Digital Storytelling.** Establishes the creative context by defining DS, the concept of collaboration, and the operational role of co-creation platforms.
- **Generative AI.** Explores the technological domain, addressing the concept, evolution, and distinction of GenAI from traditional AI, along with its specific applications in creative and educational fields.
- **AI Assistant in Co-creation.** Bridges the two domains, focusing on Human-AI co-creation in narrative generation and analyzing the potential of AI assistants as optimizing tools within the creative process.

Digital Storytelling (DS), also referred to as Digital Narrative, has emerged as a significant evolution of traditional storytelling practices by integrating multimedia elements such as text, image, audio, video, and animation within digital environments. Rooted in the long-standing tradition of oral narration, DS bridges narrative heritage with contemporary digital affordances, enabling more immersive, participatory, and interactive forms of meaning-making. Unlike linear, printed storytelling, digital storytelling is inherently interactive and web-oriented, allowing audiences to engage through comments, annotations, and other participatory mechanisms embedded within the narrative itself [3]. As a transdisciplinary field, DS increasingly incorporates computational methods and emerging technologies, positioning itself at the intersection of narrative theory, media studies, human-computer interaction, and creative practice [4].

Collaboration constitutes a foundational principle within digital storytelling ecosystems. Across disciplines, collaboration is understood as a process in which multiple agents interact continuously, share resources, and jointly construct outcomes that exceed the potential of individual contributions [5][6]. In creative contexts, collaboration functions not merely as a means of efficiency but as a driver of innovation and experimentation. This collaborative logic is operationalized through co-creation platforms, digital systems designed to facilitate collective creativity and shared authorship [7]. However, prior research highlights persistent challenges in designing platforms that effectively support collaborative creative work, including coordination difficulties, un-

even participation, and cognitive overload [8]. As a result, many co-creation environments require facilitation mechanisms, increasingly envisioned as digital partners, such as AI assistants that can guide users, structure workflows, and enhance creative efficiency.

In parallel, Generative Artificial Intelligence (GenAI) has rapidly transformed the landscape of digital content creation. Distinguished by its ability to generate novel content that resembles human-produced data, GenAI represents a qualitative shift from earlier AI paradigms [9][10]. The field has evolved from early generative models such as Generative Adversarial Networks (GANs) to large-scale multimodal systems, including large language models like ChatGPT, capable of producing coherent text, images, and video from minimal input [9]. Unlike traditional AI and machine learning approaches, which are predominantly discriminative and focused on classification or prediction tasks, GenAI is characterized by its capacity to synthesize new and original outputs by learning underlying data structures rather than merely interpreting existing data [11][12]. This creative capability positions GenAI as a critical assistive technology for storytelling and co-creation contexts.

The impact of GenAI is particularly visible in creative and educational domains, where it introduces new opportunities for personalization, ideation, and process optimization [13][14]. In educational settings, GenAI can enrich learning experiences and support teaching practices through content generation and adaptive feedback [15]. In creative workflows, its role extends beyond automation, enabling creators to explore alternative ideas, accelerate production, and experiment with narrative forms. This evolution has reframed GenAI systems as assistants or collaborative partners rather than replacements for human creators [16]. Empirical studies of content creators, for instance on platforms such as YouTube, show that GenAI is commonly used to support specific tasks, including research, summarization, script drafting, and idea generation, while creative control remains firmly with the user [17].

At the intersection of Digital Storytelling, collaboration, and GenAI lies the paradigm of Human-AI co-creation. In this model, AI systems function as active partners that complement human creativity through suggestion, scaffolding, and rapid content generation [18]. The value of this collaboration lies in its capacity to combine computational efficiency with human intentionality, judgment, and cultural sensitivity. GenAI can introduce unexpected ideas, prompt creative exploration, and support problem-solving, while users retain agency by selecting, refining, or rejecting AI-generated inputs [19]. This synergistic relationship has become central to contemporary co-creation platforms, which increasingly integrate AI-driven functionalities such as draft generation, summarization, and narrative refinement [16][20].

Despite significant advances, a review of existing AI-powered storytelling tools reveals notable gaps. Fiction-oriented tools such as Sudowrite, AI Dungeon, NovelAI, and Artflow excel in imaginative content generation but lack robust support for uploading, contextualizing, and remixing large volumes of user-generated or archival material. Conversely, rigor-focused tools such as Grammarly or Canva Magic Write prioritize polishing, factual coherence, and multimodal integration, yet do not fully support collaborative storytelling grounded in archival or documentary content. This benchmark-

ing highlights the absence of integrated solutions that combine collaborative digital storytelling, GenAI-assisted co-creation, and the enrichment of existing multimedia archives. Addressing this gap represents a key opportunity for advancing the state of the art in AI-supported storytelling platforms and confirms the relevance of developing tailored AI assistants that balance creative support with user control and contextual rigor.

Ultimately, the study seeks to ideate and develop an intelligent assistant capable of generating narratives from existing content. The core goal is to evaluate the quality and user satisfaction of this support from the creators' perspective, while ensuring that users maintain creative control over the final output.

3.1 Digital Storytelling

Digital Storytelling also known as Digital Narrative, is an evolving narrative form that combines multimedia tools, such as images, videos, and animations, to create narratives that are more engaging and interactive than conventional ones. Inspired by the tradition of oral narration, DS has evolved with the advance of digital technologies, consolidating itself as an innovative practice that connects the narrative past to the expressive possibilities of the modern era.

Defining Digital Storytelling. Digital Storytelling refers to the use of media types, including voice, image, video, music, and graphics, in the construction of narratives, namely developed for web-based contexts [3]. Differentiating itself from traditional printed storytelling, DS promotes a more direct and engaging interaction with the audience, who can participate through comments, annotations and other interactive resources, often incorporated into the digital element itself [3]. It is a transdisciplinary field of research, characterized by a growing integration of computational methods and emerging technologies in the creation of digital narratives [4].

The Concept of Collaboration. Collaboration is a structural concept in various domains of study, characterized as the essential process where multiple agents work together to achieve a common goal [5]. The value of this practice lies in its ability to potentiate results that would be unattainable through isolated efforts [6]. In the creative context, collaboration is not just an efficiency tool, but a catalyst for innovation. In summary, collaboration is defined by continuous interaction, resource sharing, and the joint construction of a final product.

Co-creation Platforms. Co-creation platforms represent a digital mechanism that uses the concept of collaboration, allowing interaction and joint work among users [7]. However, the design of systems that support collaborative creativity faces significant challenges [8]. The implementation of these platforms often requires the presence of external facilitators or support mechanisms that guide the creative process and minimize the

challenges that are common in group work. This requirement for a support agent is fundamental for this study, as it establishes the need for a partner, namely a digital one, for example an AI assistant, which can promote efficiency and guide the user in content creation.

3.2 Generative AI

GenAI is an area of AI that is distinguished by its capacity to create new content that resembles human-generated data [9]. Its rapid development in recent decades has marked the beginning of a new era in digital content production, driving rapid growth in AI-generated content [10].

The field has progressed significantly from the initial development of models, such as Generative Adversarial Networks (GANs), to the emergence of large-scale models, including ChatGPT [9]. The most notable turning point of GenAI lies in its capacity to generate complex and varied content, which can include text, image, and video, from simple inputs [9]. This capacity of generative AI to create complex and coherent content at an unprecedented rate is what positions it as a crucial assistive technology.

Defining Digital Storytelling. Distinction from Traditional AI. The distinction between traditional AI and generative AI is crucial for understanding their application in co-creation. Traditional AI and Machine Learning (ML) approaches are often focused on discriminative tasks, where the primary objective is to analyze, classify, or predict based on existing data [11]. In contrast, GenAI differentiates itself by having the capacity to synthesize new and original content that did not previously exist in the training dataset [12]. The objective of GenAI is not just to interpret data, but to learn its underlying structures to generate unique outputs. This capacity for creation, rather than mere classification or prediction, makes GenAI the ideal technology for the project.

Creative and Educational Applications. The rapid evolution of generative AI has led to its widespread adoption across various sectors, with notable implications in the creative and educational fields [13]. GenAI presents significant opportunities and challenges for education, requiring schools to develop strategies for its effective integration [14]. The capacity of GenAI to produce text, images, and other types of media enables the enrichment of the learning experience and provides additional support to the teaching process [15]. Regarding creative applications, GenAI goes beyond mere efficiency. GenAI is a transformative technology that offers new opportunities for personalization, creation, and optimization in creative and educational environments [14].

AI Assistants in Human-AI Co-creation. The evolution of GenAI has transformed its role from a simple algorithm to that of an assistant or direct partner in the content creation process [16]. This perspective positions GenAI not as a substitute but as a support tool that optimizes tasks and increases the user's creative efficiency. The primary value of these assistants lies in their ability to generate and manipulate content rapidly, which is particularly relevant in the domain of User-Generated Content [16]. Research on con-

tent creators on platforms like YouTube demonstrates that users apply GenAI to optimize specific aspects of their workflow, such as script research, title generation, summarization, and idea inspiration [17].

Technical Context and Model Benchmarking. Considering the preliminary findings [1][2] and the goal of developing an optimized assistant for the Polariscope platform, a robust technical benchmarking was conducted to determine the best AI implementation approach. Following prior research, the solution involved adding two crucial parameters for output control: Top-p, which restricts the generated vocabulary, focusing the result on the intended theme, and Repetition Penalty, which forces the model to vary the words used within the chosen theme. Large Language Models (LLMs) such as ChatGPT 4-o and Gemini were identified as the main candidates, due to their proven excellence and their robust and well-known APIs. While open models like Llama 3 were also considered, the necessary hardware cost (especially the RAM on the virtual machine) would negate any expected cost benefits.

3.3 AI in cocreation

After establishing the conceptual pillars of Digital Storytelling and Collaboration (Section 3.1) and framing the technological advancement of Generative AI (Section 3.2), this section focuses on their intersection. The objective is to analyze how AI models can be integrated into digital platforms, acting as an active assistant and partner in the story creation process. The emphasis is placed on Human-AI co-creation models and how AI can optimize the creative workflow while simultaneously ensuring user control over the final result.

Human-AI Co-creation. This type of co-creation represents a collaboration paradigm where Generative AI systems evolve beyond a simple tool, taking on the role of an active partner in the creative process [18]. This collaboration is considered the most effective way to integrate technology into creation, generating synergies that combine the machine's processing capacity with human intention and sensibility [18]. The fundamental value of GenAI in this context lies in its capacity to stimulate creativity and problem-solving. AI can suggest ideas that would otherwise not occur to users, allowing the creator to select, refine, or use these inputs to generate new solutions [19]. This intervention is viewed as a scaffolding that guides users through a series of creative stages, enhancing their productivity and the novelty of their ideas [19]. In summary, Human-AI co-creation is a model that maximizes creative potential through the combination of AI's idea generation capabilities and the user's creative control [19].

AI-powered Co-creation Platforms. Co-creation platforms have evolved significantly with the integration of GenAI, transforming into ecosystems that unite the machine's generation capacity with human creative control [16][20]. The focus is on utilizing AI to optimize collaborative creation through functionalities such as draft generation, input summarization, and improvement of narrative quality. The importance of evaluating these tools is crucial, requiring a comparative analysis of different models to determine

their effectiveness in terms of functionality and ease of use in collaboration [20]. Thus, the success of digital co-creation critically depends on the strategic choice and integration of an AI assistant that can facilitate creation while maintaining the quality of the outputs on the platform.

Benchmarking AI Co-creation in Storytelling. The analysis of the state of the art in Human-AI co-creation must be complemented with a comparative evaluation of existing tools on the market. This benchmarking is crucial for identifying gaps and positioning the development of the Polariscope assistant. In this study, AI tools were evaluated, divided into two main focuses: fiction creation and contextual rigor. Tools such as Sudowrite, AI Dungeon, NovelAI, and Artflow, although excellent for producing fiction content, revealed a critical limitation in the Polariscope context: the absence of support for uploading and remixing large volumes of user-generated content. In contrast, tools focused on rigor and support for existing work were evaluated. For example, Grammarly, while integrating text generation features, maintains its primary focus on polishing and correcting writing. Canva Magic Write was examined for its ease in generating multimodal content for social networks, demonstrating the ability to create factual outputs and integrate various media (such as photos), depending on the given prompt. Based on this benchmarking, the market demonstrates a significant gap in solutions that combine the identified needs: a collaborative storytelling platform that integrates AI as an assistive tool for creators, focused on the use and enrichment of multimedia records from archive content. This result confirms the need to develop and evaluate a solution that fills this gap in the story co-creation process.

4 Methodology

This research is anchored in the Design-Based Research (DBR) methodology. DBR is chosen for its practical, problem-solving orientation, which aligns perfectly with the goal of developing and refining the AI assistant (the artifact) to improve the creative experience. The investigation is explanatory, seeking to answer the fundamental questions of why and how the introduction of the GenAI assistant impacts the quality and ease of the co-creation process. The DBR process is inherently interventionist (testing the artifact in a real-world setting) and iterative (using successive cycles of design, evaluation, and revision to refine the assistant). Hence, the following 4 phases were defined:

- **Phase 1.** State of the Art and Conceptualization (concepts; benchmarking of related work; personas; and functional requirements).
- **Phase 2.** Prototype Development (mockups and technical integration of services based on the conceptualized framework).
- **Phase 3.** Laboratory testing and Iteration (development and application of data collection instruments for evaluation of the prototype with users, followed by corrections and optimization).
- **Phase 4.** Field trial evaluation (development and application of a triangulation of data collection instruments, followed by final corrections and optimization).

The current paper presents the outcomes of Phases 1 and 2, detailing the conceptualized framework and its technical architecture.

For the upcoming evaluation phases, participants will be selected via convenience sampling and will include students from the University of Aveiro and students and teachers from a secondary school in Aveiro. We expect that these groups have experience with digital applications to avoid having accessibility barriers for standard App usage, so the research may focus on specific functionality. Two profiles are included: common users (to assess ease of creation) and experienced creators who are already familiar with the first version of Polariscope (no AI enhanced).

A mixed-methods approach will be used for data collection, conducted across Phase 3 and Phase 4. The data collection instruments are tailored to each evaluation phase: Phase 3 will focus on quick usability and functional feedback via short qualitative sessions within focus groups to assess usability issues, workflow flaws, and acute design needs, informing immediate interface adjustments. Phase 4 will involve a user experience assessment using remote in-app questions and quantitative questionnaires applied at scale, including validated instruments like the System Usability Scale (SUS) and AttrakDiff. This will be complemented by in-depth qualitative interviews with a group of participants for triangulation, which will explore the creators' motivations, sense of ownership, and the perceived quality of the stories to provide a comprehensive understanding of the results.

5 Proposed Solution

To move beyond a purely conceptual proposition, the proposed framework establishes a concrete technical architecture for integrating Generative AI into the Polariscope platform. The solution is designed to assist creators dynamically at the point of need. The overall architecture is structured into three interdependent layers: the Client Interface (Frontend), the Context Processing Engine (Backend), and the Large Language Model Inference (Cloud) (see Figure 1):

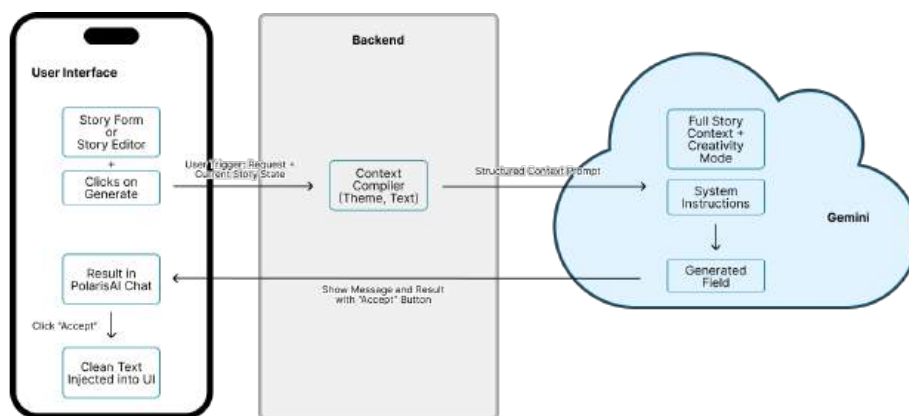


Fig. 1. Polariscope Architectural Workflow for AI-Assisted Story Co-Creation.

From a technical standpoint, the intelligent assistant (PolarisAI) is powered by the Google Gemini API, specifically utilizing the gemini-3-flash-preview model. To ensure the AI acts as a contextual co-author rather than a generic chatbot, a dedicated backend module handles strict prompt engineering. The framework integrates seamlessly into the user workflow by providing the following core capabilities as shown by Figure 2:

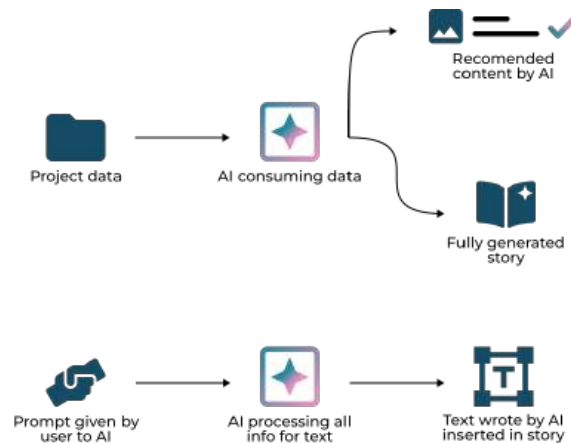


Fig. 2. Polariscope AI assistant Key Functionalities and their Workflow.

Intelligent Text Assistance. To prevent writer’s block while filling in text fields (titles, descriptions, or story paragraphs), the architecture defines two interaction paradigms. First, in-context quick buttons dynamically switch between "Generate" and "Rewrite" depending on whether a field is empty, allowing single-click assistance. Second, a dedicated chat interface allows users to insert specific instructions. When triggered, the backend dynamically compiles the user’s current progress (including the project’s name, available metadata of the contents, and previously written text blocks), into a structured context prompt. To guarantee that the AI’s output can be easily integrated into the platform’s UI, strict system instructions force the model to return responses using predefined structural tags (e.g., [[TITLE]], [[TEXT]]). A custom parser in the frontend processes these tags and presents the suggestions cleanly to the user as interactive cards with an “Accept” button that can be injectes the result directly into the story.

Content Recommendation. The AI scans the metadata of existing project content to identify assets that semantically align with the story the creator is currently writing. When the user selects the “Add Content” button, the backend matches the ongoing narrative context with the database archives, ensuring that the top multimedia contents deemed most relevant by the AI appear prioritized, bridging the gap between text generation and visual storytelling.

Full Story Generation. For projects with a substantial volume of content, able to be processed by GenAI to create a narrative, a dedicated button will be available on the

main project page. This function enables the project creator to automatically generate a complete story with minimal effort. However, this feature requires further assessment and evaluation.

This architectural framework ensures that the AI acts as a supportive layer. The prototype following these technical specifications is currently under development so that the research team can engage participants in the subsequent evaluation and optimization stages outlined in the methodology.

6 Conclusion

This paper presented a technical and conceptual framework for AI-assisted storytelling, aimed at addressing a core challenge in collaborative digital storytelling platforms: supporting users in transforming distributed multimedia contributions into coherent and meaningful narratives. Within the context of the Polariscope platform, the research explored how GenAI can be integrated as an assistive technology through a dedicated architecture to reduce barriers to story creation, such as limited time, creative confidence, and writer's block, while preserving human authorship and creative control.

The proposed solution positions AI as a creative partner within a Human-AI co-creation paradigm rather than as an autonomous content producer. The specified architecture defines a clear data flow involving context compilation, multimodal model inference via the Gemini API, and UI integration through custom parsing. Three complementary functionalities were established, targeting specific needs in the creative workflow: content recommendation, intelligent text assistance, and full story generation. By embedding AI assistance at moments of need through structured backend prompts, the framework seeks to increase efficiency, engagement, and the perceived quality of collaboratively created stories.

A central contribution of a prior assessment regarding automatic story creation [1][2] lies in the recognition that creativity in AI-generated narratives is highly context-dependent. Different story types require varying degrees of generative freedom, making adaptive control of AI parameters essential. This insight, combined with an updated benchmarking analysis of existing tools, revealed a clear gap in current solutions: the absence of platforms that effectively integrate AI-driven narrative support with the structured reuse of collective multimedia archives.

The research adopts a Design-Based Research methodology to ensure that technological development is closely aligned with iterative evaluation. With the architectural framework and initial prototype defined, future work will focus on the execution of Phase 3 and Phase 4. Controlled laboratory testing will be conducted to assess acute usability needs and workflow efficiency. Subsequently, a field trial will be conducted using real projects developed within the Polariscope platform by students of the University of Aveiro and the community of a secondary school in Aveiro. This trial will

evaluate the assistant's impact on story co-creation scenarios, providing empirical evidence of its effectiveness and receptivity to inform the final refinement of the platform.

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